

# OE-EC804B — Microwave Integrated Circuits

Semester VIII • 3L:0T:0P • 3 Credits

## Module 2 Exam Notes: Planar Transmission Lines-I (Stripline & Microstrip)

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### Contents

|          |                                                                                 |          |
|----------|---------------------------------------------------------------------------------|----------|
| <b>1</b> | <b>Stripline and Microstrip Line Fundamentals</b>                               | <b>2</b> |
| 1.1      | Stripline Geometry and Pure TEM Propagation . . . . .                           | 2        |
| 1.2      | Microstrip Line Geometry and Quasi-TEM Mode . . . . .                           | 2        |
| <b>2</b> | <b>Effective Dielectric Constant (<math>\epsilon_{eff}</math>) and Analysis</b> | <b>2</b> |
| 2.1      | Concept of Effective Dielectric Constant ( $\epsilon_{eff}$ ) . . . . .         | 2        |
| 2.2      | Guided Wavelength and Phase Velocity . . . . .                                  | 3        |
| 2.3      | Microstrip Characteristic Impedance ( $Z_0$ ) Analysis & Synthesis . . . . .    | 3        |
| 2.4      | Stripline Characteristic Impedance Formula . . . . .                            | 3        |
| <b>3</b> | <b>Losses, Dispersion, and Discontinuities in Microstrip</b>                    | <b>4</b> |
| 3.1      | Microstrip Attenuation & Loss Mechanisms . . . . .                              | 4        |
| 3.2      | Microstrip Discontinuities . . . . .                                            | 4        |

# 1 Stripline and Microstrip Line Fundamentals

Planar transmission lines form the foundational interconnections and resonant components in MIC design.

## 1.1 Stripline Geometry and Pure TEM Propagation

A **Stripline** consists of a flat strip conductor of width  $W$  and thickness  $t$  centered symmetrically inside a dielectric substrate of thickness  $b$ , bounded above and below by two parallel conducting ground planes.

### Stripline Pure TEM Propagation

Because the strip conductor is completely embedded inside a single, homogeneous dielectric medium ( $\epsilon_r$ ), the wave phase velocity is uniform throughout the cross-section. Stripline supports a **pure Transverse Electromagnetic (TEM) mode** with zero longitudinal electric ( $E_z = 0$ ) or magnetic ( $H_z = 0$ ) field components.

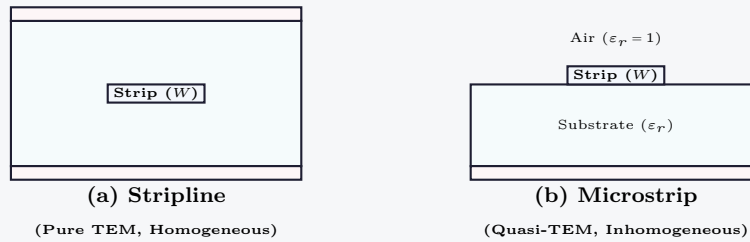
## 1.2 Microstrip Line Geometry and Quasi-TEM Mode

A **Microstrip Line** consists of a strip conductor of width  $W$  and thickness  $t$  printed on top of a dielectric substrate of thickness  $h$ , backed by a single ground plane on the bottom.

### Quasi-TEM Mode in Microstrip

Because the top surface of a microstrip line is open to air ( $\epsilon_r = 1$ ) while the lower region consists of dielectric ( $\epsilon_r > 1$ ), the electromagnetic field spans two different dielectric media. The phase velocity in air ( $c$ ) differs from that in the substrate ( $c/\sqrt{\epsilon_r}$ ). Pure TEM propagation cannot be supported; the fundamental mode is an inhomogeneous **Quasi-TEM mode** with tiny longitudinal field components ( $E_z, H_z \neq 0$ ).

### Cross-Sectional Geometries and Field Distribution: Stripline vs. Microstrip



# 2 Effective Dielectric Constant ( $\epsilon_{eff}$ ) and Analysis

## 2.1 Concept of Effective Dielectric Constant ( $\epsilon_{eff}$ )

To simplify the analysis of inhomogeneous microstrip lines, the air-dielectric boundary is replaced by a fictitious homogeneous dielectric medium extending throughout space, characterized by an **Effective Dielectric Constant** ( $\epsilon_{eff}$ ).

### Effective Dielectric Constant ( $\epsilon_{eff}$ ) Formula

The effective dielectric constant is bounded by  $1 < \epsilon_{eff} < \epsilon_r$  and is calculated using Schneider's closed-form expression:

$$\epsilon_{eff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left[ 1 + 12 \frac{h}{W} \right]^{-1/2} \quad (1)$$

where  $W$  is the microstrip width and  $h$  is the substrate thickness.

## 2.2 Guided Wavelength and Phase Velocity

The phase velocity  $v_p$  and guided wavelength  $\lambda_g$  in microstrip are given by:

$$v_p = \frac{c}{\sqrt{\epsilon_{eff}}} \quad ; \quad \lambda_g = \frac{v_p}{f} = \frac{\lambda_0}{\sqrt{\epsilon_{eff}}} \quad (2)$$

## 2.3 Microstrip Characteristic Impedance ( $Z_0$ ) Analysis & Synthesis

1. Analysis Formulas (Given  $W/h$  and  $\epsilon_r$ , find  $Z_0$ ):

- For Narrow Strips ( $W/h \leq 1$ ):

$$Z_0 = \frac{60}{\sqrt{\epsilon_{eff}}} \ln \left[ 8 \frac{h}{W} + 0.25 \frac{W}{h} \right] \quad (3)$$

- For Wide Strips ( $W/h \geq 1$ ):

$$Z_0 = \frac{120\pi}{\sqrt{\epsilon_{eff}} \left[ \frac{W}{h} + 1.393 + 0.667 \ln \left( \frac{W}{h} + 1.444 \right) \right]} \quad (4)$$

2. Synthesis Formulas (Given target  $Z_0$  and  $\epsilon_r$ , find required  $W/h$ ):

- For  $W/h < 2$ :

$$\frac{W}{h} = \frac{8e^A}{e^{2A} - 2} \quad \text{where } A = \frac{Z_0}{60} \sqrt{\frac{\epsilon_r + 1}{2}} + \frac{\epsilon_r - 1}{\epsilon_r + 1} \left( 0.23 + \frac{0.11}{\epsilon_r} \right) \quad (5)$$

- For  $W/h > 2$ :

$$\frac{W}{h} = \frac{2}{\pi} \left[ B - 1 - \ln(2B - 1) + \frac{\epsilon_r - 1}{2\epsilon_r} \left( \ln(B - 1) + 0.39 - \frac{0.61}{\epsilon_r} \right) \right] \quad (6)$$

$$\text{where } B = \frac{377\pi}{2Z_0\sqrt{\epsilon_r}}.$$

## 2.4 Stripline Characteristic Impedance Formula

For a symmetrical stripline with ground plane spacing  $b$  and effective strip width  $W_e$ :

$$Z_{0,\text{stripline}} = \frac{30\pi}{\sqrt{\epsilon_r}} \frac{b}{W_e + 0.441b} \quad \text{where } W_e = W \text{ for } t/b \rightarrow 0 \quad (7)$$

### Worked Numerical: Microstrip Design & Wavelength Calculation

**Problem:** A microstrip line is printed on an Alumina substrate ( $\epsilon_r = 9.8$ ,  $h = 0.635$  mm). If the width-to-height ratio is  $W/h = 1.0$ , calculate:

- The effective dielectric constant  $\epsilon_{eff}$ .
- The characteristic impedance  $Z_0$ .
- The guided wavelength  $\lambda_g$  at 10 GHz.

**Solution:**

- Calculate  $\epsilon_{eff}$  using  $W/h = 1.0$ :

$$\begin{aligned} \epsilon_{eff} &= \frac{9.8 + 1}{2} + \frac{9.8 - 1}{2} [1 + 12(1)]^{-1/2} \\ &= 5.4 + 4.4 [13]^{-1/2} = 5.4 + \frac{4.4}{3.6055} \\ &= 5.4 + 1.2203 = 6.6203 \end{aligned} \quad (8)$$

(b) Since  $W/h = 1.0 \leq 1$ , use the narrow strip formula for  $Z_0$ :

$$\begin{aligned} Z_0 &= \frac{60}{\sqrt{6.6203}} \ln [8(1) + 0.25(1)] \\ &= \frac{60}{2.573} \ln(8.25) = 23.319 \times 2.1102 \\ &= 49.21 \, \Omega \approx 50 \, \Omega \end{aligned} \quad (9)$$

(c) Free-space wavelength  $\lambda_0 = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{10 \times 10^9 \text{ Hz}} = 30 \text{ mm}$ .

$$\lambda_g = \frac{\lambda_0}{\sqrt{\epsilon_{eff}}} = \frac{30 \text{ mm}}{\sqrt{6.6203}} = \frac{30}{2.573} = 11.66 \text{ mm} \quad (10)$$

### 3 Losses, Dispersion, and Discontinuities in Microstrip

#### 3.1 Microstrip Attenuation & Loss Mechanisms

Total attenuation constant  $\alpha_{total} = \alpha_c + \alpha_d + \alpha_r$  (Np/m or dB/m):

- **Conductor Loss ( $\alpha_c$ ):** Skin effect causes current to crowd within skin depth  $\delta = \sqrt{\frac{1}{\pi f \mu \sigma}}$  along lower surface edges:

$$\alpha_c \approx \frac{8.686 R_s}{Z_0 W} \text{ dB/m} \quad \text{where } R_s = \sqrt{\frac{\pi f \mu}{\sigma}} \text{ (Surface Resistance)} \quad (11)$$

- **Dielectric Loss ( $\alpha_d$ ):** Caused by substrate dipole relaxation loss:

$$\alpha_d = 27.3 \frac{\epsilon_r}{\sqrt{\epsilon_{eff}}} \frac{\epsilon_{eff} - 1}{\epsilon_r - 1} \frac{\tan \delta}{\lambda_0} \text{ dB/m} \quad (12)$$

- **Radiation Loss ( $\alpha_r$ ):** At high frequencies, microstrip open boundaries act as long slot radiators emitting RF energy into space.

#### 3.2 Microstrip Discontinuities

Physical transitions in microstrip layouts alter local electric and magnetic field distributions, generating parasitic reactances.

| Discontinuity Type            | Physical Origin                                                             | Equivalent Circuit Model & Mitigation                                                                                                                                |
|-------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Open-End Line</b>          | Fringing electric fields extend past strip termination into air             | Shunt end capacitance $C_{oc}$ , equivalent to extra length<br>$\Delta l = 0.412h \frac{\epsilon_{eff} + 0.3}{\epsilon_{eff} - 0.258} \frac{W/h + 0.264}{W/h + 0.8}$ |
| <b>Step-Change in Width</b>   | Sudden width transition $W_1 \rightarrow W_2$ disrupts current distribution | Series inductance $L_s$ and shunt capacitance $C_s$ .                                                                                                                |
| <b>Right-Angle Bend (90°)</b> | Current crowding at inner corner; excess charge at outer corner             | Series inductance $L_b$ and shunt capacitance $C_b$ . Mitigated by 45° Mitred Bend ( $M = 52 + 65e^{-1.35W/h}\%$ metal removal).                                     |
| <b>T-Junction / Cross</b>     | Junction of orthogonal microstrip branches                                  | Equivalent T-network of series inductances and shunt capacitance.                                                                                                    |

## Quick Revision — Module 2

Planar Transmission Lines-I (Stripline & Microstrip) | OE-EC804B

### Key Concepts and Metrics at a Glance

|                                                         |                                                                                                                                                         |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Effective Dielectric <math>\epsilon_{eff}</math></b> | $\epsilon_{eff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left[ 1 + 12 \frac{h}{W} \right]^{-1/2} \quad (1 < \epsilon_{eff} < \epsilon_r)$ |
| <b>Phase Velocity &amp; Wavelength</b>                  | $v_p = \frac{c}{\sqrt{\epsilon_{eff}}} \quad ; \quad \lambda_g = \frac{\lambda_0}{\sqrt{\epsilon_{eff}}}$                                               |
| <b>Stripline Propagation</b>                            | Pure TEM mode ( $E_z = 0, H_z = 0$ ) due to symmetrical homogeneous dielectric.                                                                         |
| <b>Microstrip Propagation</b>                           | Inhomogeneous Quasi-TEM mode ( $E_z, H_z \neq 0$ ) due to air-substrate boundary.                                                                       |
| <b>Mitred Bend Optimal Cut</b>                          | $M = \frac{x}{d\sqrt{2}} \times 100\% = 52 + 65e^{-1.35W/h}\%$ removes $90^\circ$ corner capacitance.                                                   |

### One-Line Definitions

- **Quasi-TEM Mode:** Hybrid electromagnetic wave in non-homogeneous media possessing tiny longitudinal field components.
- **Fringing Fields:** Electric field lines extending into the air region surrounding microstrip edges.
- **Mitred Bend:** A  $45^\circ$  chamfered microstrip bend removing excess corner capacitance to maintain 50  $\Omega$  matching.
- **Microstrip Dispersion:** Frequency-dependent variation of  $\epsilon_{eff}(f)$  causing phase velocity to decrease at high frequencies.

### Frequently Asked / Viva Questions

**Q1. Why does a microstrip line support a Quasi-TEM mode instead of a pure TEM mode?**

Because the electromagnetic field exists in two different media (dielectric substrate and air), causing phase velocity mismatch that induces small longitudinal  $E_z$  and  $H_z$  components.

**Q2. Why does a stripline support a pure TEM mode?**

Because the conductor is completely enclosed inside a single homogeneous dielectric bounded by two ground planes.

**Q3. What is the Effective Dielectric Constant ( $\epsilon_{eff}$ ) of a microstrip?**

A weighted average dielectric constant of air and substrate representing a fictitious homogeneous medium supporting the same wave velocity.

**Q4. How does characteristic impedance ( $Z_0$ ) vary with strip width  $W$ ?**

As strip width  $W$  increases, capacitance increases and characteristic impedance  $Z_0$  decreases ( $Z_0 \propto 1/W$ ).

**Q5. Explain the physical significance of an open-end microstrip discontinuity.**

Fringing fields at the open termination store electric energy, behaving as a shunt capacitor  $C_{oc}$  that increases the line's effective length by  $\Delta l$ .

**Q6. Why are right-angle microstrip bends mitred (chamfered)?**

A  $90^\circ$  sharp bend creates excess capacitance at the outer corner. Mitring removes 50% to 60% of the corner metal, eliminating reflections.

**Q7. What are the main loss mechanisms in microstrip lines?**

**Frequently Asked / Viva Questions (contd.)**

Conductor loss  $\alpha_c$  (skin effect), dielectric loss  $\alpha_d$  (substrate dissipation), and radiation loss  $\alpha_r$  (open boundary power leakage).

**Q8. Differentiate between Stripline and Microstrip in terms of isolation.**

Stripline provides superior shielding against crosstalk due to dual ground planes. Microstrip suffers higher radiation loss due to its open top boundary.

**Q9. How does dispersion affect microstrip performance at high frequencies?**

As frequency increases, fields become more confined inside the substrate, increasing  $\varepsilon_{eff}(f)$  towards  $\varepsilon_r$  and causing dispersion (signal distortion).

**Q10. What is the typical value of  $\varepsilon_{eff}$  relative to  $\varepsilon_r$ ?**

$\varepsilon_{eff}$  is strictly bounded by  $1 < \varepsilon_{eff} < \varepsilon_r$ , typically around  $0.65\varepsilon_r$  for standard microstrips.